

holodev

This repository contains examples and experiments using HoloOcean (AUV/USV simulator) with EKF-based sensor fusion and small utility scripts. The main code for filtering and experiments lives in the `klamaning/` folder.

Project highlights

- **EKF experiments:** Multi-sensor EKF fusion (IMU, DVL, Depth, Acoustic) with visualization.
- **Interactive control:** Keyboard thruster control examples for manually driving agents in-sim.
- **Utilities & examples:** Helper functions for uncertainty visualization and various HoloOcean examples.

Contents

- `klamaning/`: main experiments and utilities
 - `currents.py` — full EKF demo comparing 4 sensor configurations (IMU / IMU+DVL / IMU+DVL+Depth / IMU+DVL+Depth+Acoustic). Produces trajectory plots and uncertainty visualizations.
 - `compare.py` — simpler IMU vs IMU+DVL comparison with RMSE and plots.
 - `imu.py` — small example that reads IMU, logs acceleration history and saves `accel_history.csv` + `accel_plot.png`.
 - `uncertainty_utils.py` — helpers for covariance ellipses/ellipsoids used by plotting code.
 - `ekf.ipynb`, `imu_dvl.ipynb` — notebooks for interactive analysis.
- Top-level scripts: quick examples and scenarios
 - `examples.py` — assorted HoloOcean usage examples and dynamics demos.
 - `currents.py`, `klamaning.py`, `manual.py`, `try.py` — small demos for currents, keyboard control, acoustic messages.
 - `four_ranges.py`, `accel_history.csv` — other utilities and sample data.

Requirements

- **Python:** 3.8+ recommended
- **Key packages:** `holocean`, `numpy`, `matplotlib`, `pynput`.
- Install (example):

```
pip install numpy matplotlib pynput
# Install HoloOcean per its docs (may be via pip or local package/setup
specific to your environment)
```

Note: `holocean` is an external simulator — follow its installation and scenario setup instructions before running the examples.

Quick start

- Run the main EKF comparison (full demo):

```
python klamaning/currents.py
```

- Run the lighter comparison (IMU vs IMU+DVL):

```
python klamaning/compare.py
```

- Collect IMU acceleration history + plot:

```
python klamaning/imu.py  
# produces `accel_history.csv` and `accel_plot.png`
```

Keyboard controls (common mapping)

- **Forward / Back:** `i` / `k` (applies forward/back thrust)
- **Yaw left / right:** `j` / `l`
- **Up / Down (vertical):** `w` / `s`
- **Roll left / right:** `a` / `d`
- **Quit example loops:** typically `q` (varies by script)

Configuration tips

- Many scripts use the `SCENARIO_NAME` or hard-coded scenario like `blue_rov` or `usv_auv`. Edit the top of the script if your scenario name differs.
- Noise / filter parameters are tunable constants near the top of the EKF scripts (`Q_*`, `P_*`, `*_STD` values).

Outputs

- `accel_history.csv` — example saved in the repo (produced by `klamaning/imu.py` when run).
- Plots — scripts call `matplotlib.pyplot.show()` and may save figures (e.g., `accel_plot.png`).

Notebooks

- `klamaning/ekf.ipynb` and `klamaning/imu_dvl.ipynb` are available for interactive exploration and visualizing results.

Contributing / next steps

- If you want, I can:
 - run a smoke test (if `holocean` is installed here),
 - add a `requirements.txt` or `pyproject.toml`,
 - standardize scenario configuration into a single config file.

If you'd like any of those, tell me which and I will proceed.